ALGEBRAIC MORPHISMS WITH REPRESENTABLE TARGET DIAGONAL HAVE REPRESENTABLE SOURCE DIAGONAL

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Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Representable diagonal criterion. algebraic-lemma-representable-diagonal (cross-chapter, named). Reduces to Isom(x_1, x_2) being an algebraic space.
- Representable over algebraic space. bootstrap-lemma-representable-by-spaces-over-space (cross-chapter, named). Reduces to f : Isom(x_1,x_2) -> Isom(y_1,y_2) being representable.

Source reference. Original reference 045G: focused-seeds/geometry/S9-crit-diagonals-and-algebraic-morphisms/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02YS: focused-seeds/geometry/S9-crit-diagonals-and-algebraic-morphisms/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $F : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of stacks in groupoids over $(Sch/S)_{fppf}$. If F is algebraic and $\Delta : \mathcal{Y} \to \mathcal{Y} \times \mathcal{Y}$ is representable by algebraic spaces, then $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces.*

Proof. Assume F is algebraic and $\Delta : \mathcal{Y} \to \mathcal{Y} \times \mathcal{Y}$ is representable by algebraic spaces. Take a scheme U over S and two objects x_1, x_2 of $\mathcal{X}$ over U. We have to show that $Isom(x_1, x_2)$ is an algebraic space over U, see Algebraic Stacks, Lemma 045G. Set $y_i = F(x_i)$. We have a morphism of sheaves of sets

$$f : Isom(x_1, x_2) \to Isom(y_1, y_2)$$

and the target is an algebraic space by assumption. Thus it suffices to show that f is representable by algebraic spaces, see Bootstrap, Lemma 02YS. Thus we can choose a scheme V over U and an isomorphism $\beta : y_{1,V} \to y_{2,V}$ and we have to show the functor

$$(Sch/V)_{fppf} \to Sets, \quad T/V \mapsto \{\alpha : x_{1,T} \to x_{2,T} \text{ in } \mathcal{X}_T \mid F(\alpha) = \beta|_T\}$$

is an algebraic space. Consider the objects $z_1 = (V, x_{1,V}, \mathrm{id})$ and $z_2 = (V, x_{2,V}, \beta)$ of

$$\mathcal{Z} = (Sch/V)_{fppf} \times_{y_{1,V}, \mathcal{Y}} \mathcal{X}$$

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Then it is straightforward to verify that the functor above is equal to $Isom(z_1, z_2)$ on $(Sch/V)_{fppf}$. Hence this is an algebraic space by our assumption that F is algebraic (and the definition of algebraic stacks). $\square$